

UAC Program Predictive Capacity Forecast

Executive Summary

1. The Problem

The UAC Program currently relies on historical reporting to understand the number of children in federal care. This means staffing, shelter capacity, and placement planning decisions are often made after demand has already changed rather than in advance. This project addresses that gap by forecasting future care demand before it occurs, allowing planners to prepare resources proactively.

2. What We Predicted

We built forecasting models to predict the number of children in HHS Care and future discharge demand, looking 7 to 14 days into the future. Five different forecasting approaches were evaluated, ranging from simple historical forecasting techniques to advanced machine-learning models. The most reliable model was selected based on its performance on recent unseen data.

3. Key Finding

Our analysis identified the **Net Pressure Indicator** as an important early warning signal.

Net Pressure = Children Transferred into HHS Care – Children Discharged from HHS Care

When transfers consistently exceed discharges, operational pressure increases and future capacity strain becomes more likely. Monitoring this indicator provides planners with advance visibility into potential resource challenges before they impact shelter operations.

4. Recommendation

We recommend that HHS planners increase monitoring of staffing levels, shelter capacity, and placement resources whenever the Net Pressure indicator remains positive for multiple consecutive days. This proactive trigger point would allow operational decisions to be made before capacity constraints occur rather than during a surge event.

5. Model Confidence

Our best-performing model forecasts the number of children in HHS Care within approximately **7 children of the actual value on average** over a short-term forecast horizon. This level of accuracy is sufficient to support operational planning decisions related to staffing, shelter management, and resource allocation, while recognizing that forecast precision naturally decreases further into the future.

Business Impact

The forecasting framework transforms historical operational reporting into predictive intelligence. Instead of reacting to capacity challenges after they occur, planners can anticipate future demand, prepare resources in advance, and improve operational readiness. This supports better staffing decisions, more effective shelter utilization, and improved outcomes for children within the UAC Program.

Operational Trend

Historical analysis indicates that the number of children in HHS Care has followed a declining trend since its peak in early 2024. This reduction has lowered immediate capacity pressure; however, future surges remain possible and require continuous monitoring.

Early Warning Capability

The forecasting framework provides advance visibility into future care demand, enabling planners to identify potential resource constraints before they affect shelter operations. This allows more time for staffing adjustments and operational preparation.

Resource Planning Impact

Forecast outputs can support data-driven decisions related to staffing allocation, shelter utilization, medical resource planning, and placement operations. Proactive planning reduces the likelihood of emergency responses during periods of increased demand.

Final Business Impact

By combining care-load forecasting with Net Pressure monitoring, the UAC Program can transition from reactive operational management to proactive capacity planning. This improves resource utilization, strengthens operational readiness, and enhances outcomes for children under federal care.

Executive Takeaway

The forecasting model provides reliable short-term predictions and an actionable early-warning indicator, enabling HHS stakeholders to make informed operational decisions before capacity challenges emerge.